MultiLinguahah : A New Unsupervised Multilingual Acoustic Laughter Segmentation Method

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Abstract

Laughter is a social non-vocalization that is universal across cultures and languages, and is crucial for human communication, including social bonding and communication signaling. However, detecting laughter in audio is a challenging task, and segmenting it is even more difficult. Currently, Machine Learning methods generally rely on costly manual annotation, and their datasets are mostly based on English contexts. Thus, we propose an unsupervised multilingual method that sets up the laughter segmentation task as an anomaly detection of energy-based segmented audio sequences. Our method applies an Isolation Forest on audio representations learned from BYOL-A encoder. We compare our method with several state-of-the-art laughter detection algorithms on four datasets, including stand-up comedy, sitcoms, and general short audio from AudioSet. Our results show that state-of-the-art methods are not optimized for multilingual contexts, while our method outperforms them in non-English settings.

Index Terms: laughter, audio segmentation, computational paralinguistics

1. Introduction

Laughter is ever-present in human interactions, playing an important part in human-human communication, acting also as a tool for social bonding [1][2]. It is inherently social, as it not only communicates one's internal state but also helps to propagate this state to other listeners [3]. It can express joy, relief, or success, but also appears during embarrassment, anger, or even sadness [4].

Laughter has been the object of study in domains such as psychology [5], linguistics [6], or computer science [7, 8]. Laughter is useful for a broad variety of different tasks crucial to understand human communication and for building Socially Interactive Agents (SIAs; [9]): from emotion recognition to humor extraction [10, 8] and eventually natural speech generation [11]. This makes its automatic detection important.

One of the perspectives from which laughter can be studied is its acoustic structure. Recent research suggests that this structure is universal, meaning that laughter exhibits similar acoustic patterns across languages [12], impulsing the need for general language and domain-agnostic systems detecting it.

Deep Learning (DL) algorithms are robust methods for automatic laughter detection and laughter segmentation, where they aim at detecting the beginning and end of laughter in an audio file. But the most efficient algorithms to tackle this task are DL-based [13, 14, 15] and generally require a high quantity of manual annotation. Requiring the exact timestamps is a meticulous task that is very time-consuming, making unsupervised modelling very useful in this case.

Recent works, when not relying on manual annotations, proposed to annotate laughter using automatic methods such as using the transcripts [10], or artificially adding laughter to non-laughter segments [13]. The former does not give the exact position, while the latter allows for applying data-augmentation techniques right on the laughter and knowing precisely where the exact laughter starts and ends in the new audio. However, naturally occurring examples are more diverse than artificially created data, as they can drift from the original data distribution [16, 17, 18]. For instance, unsupervised methods such as FunnyNet [19] are evaluated on datasets like *Friends*, a sitcom in which speech is dominant, background noise is limited, and the sound mixing is professionally controlled. However, this setting does not reflect real-world multilingual data collected from diverse countries and acoustic environments. In such data, background music, environmental noise, and recording conditions vary considerably. In-the-wild stand-up comedy offers the type of diverse data containing challenges that can face acoustic laughter segmentation systems [20]: diverse audio capture conditions, as well as laughter types and languages. We manually annotate test data in various languages to validate our results.

In this paper, we propose an unsupervised acoustic laughter segmentation model, which performs well for different domains and languages. We show that state-of-the-art methods, if generally better on US English, are not consistently performant across languages and domains. Overall, the main contributions of this paper can be summarized as follows: *(i)* a new unsupervised method for laughter segmentation based on energy-based segmentation and isolation forests, *(ii)* applied across languages and domains through publicly available annotated datasets and new laughter annotations, *(iii)* analysed with respect to the laughter duration. Our code is available online: `https://tinyurl.com/Multilinguahah-Interspeech26`.

2. MultiLinguahah : Acoustic Laughter Segmentation

The proposed method is composed of several steps. An overview is shown in Figure 1.

2.1. Voice Removal

The first step of our approach consists of removing the speech from the audio signal, in order to retain the background, including laughter, music, and environmental sounds.

In order to isolate the human voices from the rest of the audio, we apply an off-the-shelf Audio Source Separation model $\mathcal{S}$ to the raw audio. This model allows for the separation of speech signals from non-vocal acoustic interference. In particu-

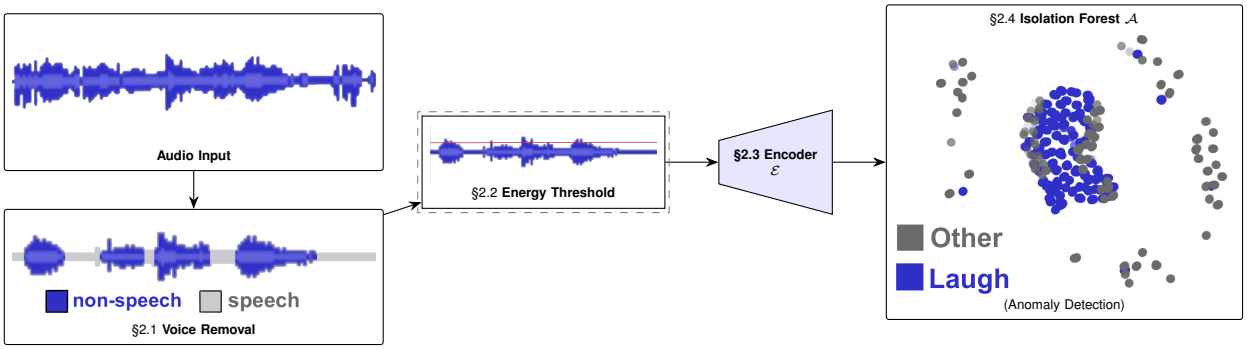

Figure 1: *We first remove the voices from the laughter through channel subtraction or audio source separation (§2.1), second then we segment the audio into events using an energy-based threshold (§2.2), third we encode the audio using a pre-trained model (§2.3), and finally we detect laughter using an anomaly detection algorithm based on Isolation Forest (§2.4).*

lar, we used a basic deep learning model based on a densely connected Convolutional Neural Network architecture [21] specifically designed for vocal signal separation [22].

In TV shows that have studio-recorded audio, laughter comes from the crowd and not from the interactants themselves. It is possible to focus only on the non-voice audio by subtracting the two audio channels. This technique was used on the Friend dataset, following the protocol of [19].

2.2. Energy-based Audio Segmentation

Once the voice has been separated from the other sources, the audio is now composed of several non-speech events that can be music, ambient sounds, or laughter. We use an energy-based peak detector [1] to find the beginning and end times of every event. Using the waveform energy as a threshold, these events are removed, retaining only the non-speech audio. The threshold was arbitrarily chosen so that slight background noise was not included in the non-silence segments. Using a lower threshold allows for more noisy events to be included, for example, if detecting very weak laughter is interesting.

2.3. Audio Encoding

Once the non-speech audio events have been extracted, a pre-trained encoder $\mathcal{E}$ was used to transform these events into vectors. For this purpose, we relied on BYOL-A [23], a self-supervised learning method for audio that learns general-purpose representations without requiring labelled data, and that has been shown useful for non-semantic speech tasks [24]. This model is initialized with weights obtained from self-supervised pretrained on the balanced and unbalanced training splits of AudioSet [25], comprising 1.963.807 audio segments (aprox. 5.455 hours), as well as the dataset FSD50K [26], which contains 40.966 audio (80 hours).

Furthermore, to perform domain adaptation, we incorporate the unlabelled training split of the target dataset into the self-supervised pre-training stage.

2.4. Anomaly Detection

Finally, in order to separate the vectors of the laughter events from the other ones, we utilize an unsupervised algorithm for anomaly detection $\mathcal{A}$. Isolation Forest [27] model was used to isolate outliers by recursively partitioning data using random feature splits. Laughter shows consistent acoustic characteristics across languages, whereas background music and other noises differ and are treated as anomalies by the model.

[1] https://github.com/amsehili/auditok

3. Experiments and Results

3.1. Datasets for Evaluation

We are validating and comparing models on a selection of 4 datasets containing laughter from various domains (in-the-wild, studio-recorded, and artificially created).

StandUp4AI [28] dataset consists of 3,617 stand-up comedy videos spanning 7 languages. It includes audience laughter annotations, capturing performances from comedians in diverse linguistic and cultural contexts. To construct the dataset, the authors collected a targeted set of stand-up comedy videos from online platforms. We added new annotations to this dataset in US English, Canadian French, and Latin American Spanish. We used the test part of the dataset, composed of 100 videos, for 8.53 hours of audio and 3.453 laughter events.

AudioSet [29] is a large-scale dataset of audio segments, each approximately 10 seconds long, sourced from **YouTube** videos and labelled with over 500 audio event categories. The dataset features diverse acoustic environments and recording qualities, covering a wide range of sound types, for 5.8 thousand hours. Laughter has been added to the audio sequences, making the timestamps known but the data artificial. The test part of the dataset was used, comprising 724 available videos and a total of 1,252 annotated laughter instances.

Friends [30] dataset is from the homonym sitcom, and consists of all 25 episodes from its third season, each approximately 23 minutes long, totalling around 10 hours of audio-visual content. The test set consists of the final 5 episodes (episodes 21–25). Within this test partition, laughter events have been manually annotated, identifying a total of 924 distinct instances of laughter.

Kuznetsova [31] is a bilingual dataset containing stand-up videos in English and Russian. The Russian subset includes 46 videos (17 hours) from 8 YouTube channels, mostly from a Vladivostok-based club. The English subset includes 56 videos (20 hours) from the largest stand-up YouTube channel. In our experiments, we focus on the official test set that has been man-

ually annotated by the original authors. This test partition comprises 10 videos in total, five in Russian (RU) and five in English (US EN). The combined duration of this evaluation set is 1.18 hours, containing a total of 617 annotated laughter instances.

3.2. Models for Evaluation

We compare our approach with three baseline models:

Gillick et al. [14] model is ResNet-based model learned in a supervised way, using classical data-augmentation audio techniques (pitch-shifting, time-stretching, and artificial reverberation) at the level of the audio. Data-augmentation allows the model to take advantage of the SwitchBoard annotated data and generalize to in-the-wild laughter detection. We follow the protocol of the authors to use the model for laughter segmentation (frame-level detection).

Omine et al. [13] model is a fine-tuning of wav2vec 2.0, trained in a supervised way on a large amount of data by randomly synthesizing data-augmented laughter in audio data of various sound qualities and from various recording environments. Using a data augmentation technique, laughter samples were inserted into the Non-laughter audio to create realistic training examples. Non-laughter audio was drawn from the Spotify Podcast Dataset [32] and AudioSet [29], and laughter samples from VocalSound [33] and Laughterscape [11].

Liu et al. [19] model is an unsupervised baseline method [19]. This baseline applies a similar methodology: it subtracts the channel to remove the voice and uses an energy-based peak detector to detect events. However, instead of employing an anomaly detection, it applies a K-means clustering algorithm to group the latent representations. For laughter detection, all clusters are retained except the smallest ones, which are assumed to correspond to non-laughter segments. With respect to the protocol, we used the same source separator as the one applied in our method when processing in-the-wild audio.

Finally, we also compare them with a mix of our model and Omine's model, showing that they can complement each other.

3.3. Evaluations Metrics

For each model, we evaluate the segmentation of laughter intervals using the Intersection over Union (IoU) metric with two thresholds: 0.3 to know how well the model detects laughter intervals, and 0.7 to evaluate the temporal segmentation. Based on these thresholds, we compute Recall and F1-score. Recall measures the proportion of ground-truth laughter segments correctly detected by the model. The F1-score is the harmonic mean of Precision and Recall.

3.4. Experimental Settings

Experiments were implemented in PyTorch and scikit-learn [34], and all computations were performed on an NVIDIA GeForce RTX 2080. For the audio encoder $\mathcal{A}$, we rely on the publicly available BYOL-A.[2] The second pre-trained process was run with a batch size of 128, a learning rate of 0.0001, for 100 epochs, using a seed of 42. The Isolation Forest was configured with the contamination parameter set to *auto*, which allows the model to calculate the anomaly threshold based on the intrinsic distribution of the data rather than a fixed percentage.

[2] https://github.com/nttcslab/byol-a

Lang	Domain	Method	F1	
			IoU=0.3	IoU=0.7
US EN	Stand-up	Gillick et al. [14]	0.456	0.134
		Omine et al. [13]	**0.679**	**0.356**
		Liu et al. [19]	0.447	0.145
		MultiLinguahah	0.506	0.176
		Omine+MultiLinguahah	0.670	0.325
	TV Show	Gillick et al. [14]	0.646	0.197
		Omine et al. [13]	0.189	0.054
		Liu et al. [19]	0.878	0.503
		MultiLinguahah	**0.910**	**0.735**
		Omine+MultiLinguahah	0.848	0.682
	Youtube	Gillick et al. [14]	0.544	0.220
		Omine et al. [13]	0.555	**0.238**
		Liu et al. [19]	0.362	0.066
		MultiLinguahah	0.315	0.087
		Omine+MultiLinguahah	**0.656**	0.206
UK EN	Stand-up	Gillick et al. [14]	0.565	0.132
		Omine et al. [13]	0.626	0.294
		Liu et al. [19]	0.733	0.394
		MultiLinguahah	0.736	0.398
		Omine+MultiLinguahah	**0.756**	**0.403**
ES	Stand-up	Gillick et al. [14]	0.294	0.076
		Omine et al. [13]	0.361	0.120
		Liu et al. [19]	0.654	0.305
		MultiLinguahah	0.649	**0.306**
		Omine+MultiLinguahah	**0.676**	0.303
Lat. ES	Stand-up	Gillick et al. [14]	0.245	0.031
		Omine et al. [13]	0.332	0.133
		Liu et al. [19]	0.572	0.187
		MultiLinguahah	0.587	0.193
		Omine+MultiLinguahah	**0.609**	**0.205**
FR	Stand-up	Gillick et al. [14]	0.149	0.009
		Omine et al. [13]	0.257	0.125
		Liu et al. [19]	0.461	0.217
		MultiLinguahah	0.543	0.264
		Omine+MultiLinguahah	**0.567**	**0.286**
Can. FR	Stand-up	Gillick et al. [14]	0.144	0.024
		Omine et al. [13]	0.237	0.107
		Liu et al. [19]	0.478	0.170
		MultiLinguahah	0.485	0.173
		Omine+MultiLinguahah	**0.521**	**0.204**
PT	Stand-up	Gillick et al. [14]	0.237	0.045
		Omine et al. [13]	0.210	0.057
		Liu et al. [19]	**0.402**	**0.179**
		MultiLinguahah	0.393	0.169
		Omine+MultiLinguahah	0.395	0.167
IT	Stand-up	Gillick et al. [14]	0.130	0.012
		Omine et al. [13]	0.391	0.157
		Liu et al. [19]	0.402	0.195
		MultiLinguahah	0.507	**0.257**
		Omine+MultiLinguahah	**0.545**	0.256
CS	Stand-up	Gillick et al. [14]	0.439	0.105
		Omine et al. [13]	0.570	0.272
		Liu et al. [19]	0.438	0.232
		MultiLinguahah	0.585	0.301
		Omine+MultiLinguahah	**0.638**	**0.321**
HU	Stand-up	Gillick et al. [14]	0.578	0.208
		Omine et al. [13]	0.706	0.376
		Liu et al. [19]	0.429	0.281
		MultiLinguahah	0.796	**0.501**
		Omine+MultiLinguahah	**0.825**	0.492
RU	Stand-up	Gillick et al. [14]	0.240	0.066
		Omine et al. [13]	0.443	0.199
		Liu et al. [19]	0.309	0.143
		MultiLinguahah	0.438	0.209
		Omine+MultiLinguahah	**0.570**	**0.254**

Table 1: *Results of the models in different languages, dialects, and domains.*

3.5. Results and Analysis

3.5.1. Global Results

Table 1 reports performance across languages, dialects, and domains. Overall, while the current state of the art from Omine et al. [13] remains highly competitive in US English domains, our approach demonstrates more stable and superior performance across languages and diverse conditions.

In US English, Omine et al.'s model achieves the best results overall, in particular on Stand-up and YouTube videos, confirming its strong performance in in-domain English settings. However, F1 drops substantially on TV Shows, even when using a different probability threshold. As we discuss in Section 3.5.3, this degradation may be explained by the prevalence of longer laughter segments in TV shows, which differ from the laughter patterns observed in stand-up or online contexts.

In contrast, our method consistently outperforms competing approaches in non-English settings, achieving the best results across Spanish, French, Italian, Czech, Hungarian, and Russian. This suggests better cross-lingual robustness and generalization beyond English-centric training conditions. We believe this is due to the backbone of Omine et al.'s model, as it is a model trained for English Audio Speech Recognition (ASR) and not for Non-Semantic Speech tasks. Finally, it is notable that Liu et al.'s model [19] obtains the strongest results for PT, which may be related to specific recording conditions and acoustic characteristics of the dataset.

3.5.2. Impact of the Audio Encoder

Table 2 reports the average performance per domain for various audio encoders. Basically, it shows that MultiLinguahah is not dependent on the encoder, as it performs similarly with several distinct models. On Stand-up, wav2clip [35] and BYOL-A [23] perform very similarly, with BYOL-A obtaining a slightly higher F1 at IoU=0.3, while wav2clip is marginally better at IoU=0.7. On TV Shows and YouTube, BYOL-A clearly outperforms wav2clip at both overlap thresholds, suggesting that self-supervised audio representations transfer particularly well to TV show data.

Dataset	**Encoder**	**F1**	
		IoU=0.3	**IoU=0.7**
Stand-up	wav2clip [35]	0.582	0.270
	BYOL-A [23]	0.584	0.269
TV Show	wav2clip [35]	0.890	0.706
	BYOL-A [23]	0.910	0.735
Youtube	wav2clip [35]	0.257	0.063
	BYOL-A [23]	0.315	0.087

Table 2: *Average performance per dataset with different audio encoders using our approach.*

3.5.3. Laughter Length Related Performances

Figure 2 shows the F1 score of MultiLinguahah vs Omine et al.'s method on the Standup4AI dataset, with respect to the laughter duration. It is possible to see that the proposed method outperforms the baseline, in particular when the laughter events last longer. Once again, we believe that the pretrained ASR backbone used by Omine et al. works against it. Indeed, as soon as the laughter is too long or the speech is not that of the training set, the model is faced with Out-of-distribution data that can disrupt its predictions.

Furthermore, our approach outperforms [19] across all laugh intervals because it better handles unpredictable noise. In noisy environments, outliers are random, which causes cluster-based strategies to fail. Because laughter shares a universal acoustic phenomenon across languages, our method succeeds by focusing on these universal patterns rather than environment-specific patterns. This allows our model to maintain precision even when the background noise is inconsistent.

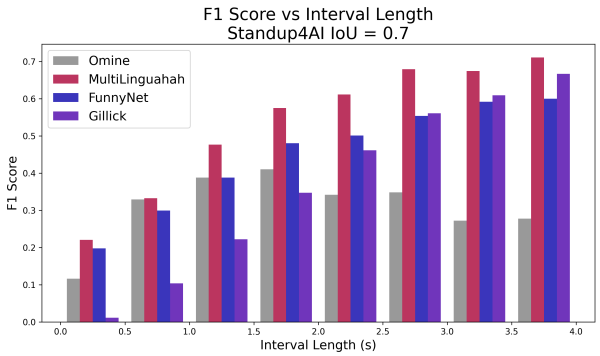

Figure 2: *Comparison of F1-scores of the proposed method against three baseline models relative to laughter duration using a temporal IoU threshold of 0.7.*

4. Conclusion

We introduced MultiLinguahah, an unsupervised multilingual method for acoustic laughter segmentation that frames the task as anomaly detection over energy-based segmented audio sequences. By combining a BYOL-A audio encoder with an Isolation Forest, our approach requires no labeled data and generalizes across languages and domains. Our experiments on four datasets covering stand-up comedy, TV shows, and YouTube content demonstrate that existing supervised methods, while competitive in US English, degrade significantly in multilingual settings. We attribute this to the ASR-oriented pretraining of models such as Omine et al., which introduces a language bias that is particularly harmful for non-English laughter. Thus, our approach is more suitable since laughter shares similar acoustic phenomena across all languages. This allows our model to focus on these universal patterns rather than environment-specific noise. In contrast, MultiLinguahah achieves consistent improvements in non-English languages, highlighting the value of non-semantic audio representations for this task. We further showed that our method is especially effective for long laughter segments, a pattern prevalent in TV shows and group laughter scenarios, and that combining MultiLinguahah with Omine et al.'s model yields complementary gains across several settings. Future work could explore adapting the energy-based segmentation to noisier environments, investigating other anomaly detection strategies, and extending the evaluation to more typologically diverse languages.

5. References

[1] R. I. M. Dunbar, R. Baron, A. Frangou, E. Pearce, E. J. C. Van Leeuwen, J. Stow, G. Partridge, I. MacDonald, V. Barra, and M. Van Vugt, "Social laughter is correlated with an elevated pain

threshold," *Proceedings of the Royal Society B: Biological Sciences*, vol. 279, no. 1731, pp. 1161–1167, 2012.

[2] R. R. Provine and K. Emmorey, "Laughter among deaf signers," *Journal of Deaf Studies and Deaf Education*, vol. 11, no. 4, pp. 403–409, 2006.

[3] R. A. MARTIN, "The social psychology of humor," *The Psychology of Humor: An integrative approach*, pp. 1–208, 2007.

[4] P. Glenn, *Laughter in interaction*. Cambridge University Press, 2003, vol. 18.

[5] A. Wood, S. Sievert, and J. Martin, "Semantic similarity of social functional smiles and laughter," *Journal of Nonverbal Behavior*, vol. 46, no. 4, 2022.

[6] J. Ginzburg, C. Mazzocconi, and Y. Tian, "Laughter as language," *Glossa: a journal of general linguistics*, vol. 5, no. 1, 2020.

[7] S. Dupont, H. Çakmak, W. Curran, T. Dutoit, J. Hofmann, G. McKeown, O. Pietquin, T. Platt, W. Ruch, and J. Urbain, "Laughter research: a review of the ilhaire project," in *Toward Robotic Socially Be-lievable Behaving Systems-Volume I: Modeling Emo-tions*, 2016, pp. 147–181.

[8] L. Hyun, K. Sung-Bin, S. Han, Y. Yu, and T. H. Oh, "SMILE: Multimodal Dataset for Understanding Laughter with Language Models," *Findings of the Association for Computational Linguistics: NAACL 2024 - Findings*, pp. 1149–1167, 2024.

[9] B. Lugrin, C. Pelachaud, and D. Traum, *The handbook on socially interactive agents: 20 years of research on embodied conversational agents, intelligent virtual agents, and social robotics volume 2: interactivity, platforms, application*. ACM, 2022.

[10] M. K. Hasan, W. Rahman, A. Zadeh, J. Zhong, M. I. Tanveer, L.-P. Morency, Mohammed, and Hoque, "UR-FUNNY: A Multimodal Language Dataset for Understanding Humor," in *EMNLP-IJCNLP*, 2019. [Online]. Available: http://arxiv.org/abs/1904.06618

[11] D. Xin, S. Takamichi, A. Morimatsu, and H. Saruwatari, "Laughter synthesis using pseudo phonetic tokens with a large-scale in-the-wild laughter corpus," in *Proc. Interspeech*, 2023.

[12] G. A. Bryant and C. M. Bainbridge, "Laughter and culture," *Philosophical Transactions of the Royal Society B*, vol. 377, no. 1863, p. 20210179, 2022.

[13] T. Omine, K. Akita, and R. Tsuruno, "Robust Laughter Segmentation with Automatic Diverse Data Synthesis," in *Interspeech*, no. September, 2024, pp. 4748–4752.

[14] J. Gillick, W. Deng, K. Ryokai, and D. Bamman, "Robust Laughter Detection in Noisy Environments," in *Proceedings of the Annual Conference of the International Speech Communication Association, INTERSPEECH*, vol. 1. International Speech Communication Association, 2021, pp. 736–740.

[15] T. Matsuda and Y. Arimoto, "Detection of laughter and screaming using the attention and ctc models," in *Proceedings of INTERSPEECH*, 2023, pp. 1025–1029.

[16] T. Naous, M. J. Ryan, A. Ritter, and W. Xu, "Having Beer after Prayer? Measuring Cultural Bias in Large Language Models," *ACL*, 2024. [Online]. Available: http://arxiv.org/abs/2305.14456

[17] T. Quiroga, F. Bravo-Marquez, and V. Barriere, "Adapting Bias Evaluation to Domain Contexts using Generative Models," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Suzhou, China: Association for Computational Linguistics, 11 2025, pp. 28 055–28 066. [Online]. Available: https://aclanthology.org/2025.emnlp-main.1424/

[18] V. Barriere and S. Cifuentes, "A Study of Nationality Bias in Names and Perplexity using Off-the-Shelf Affect-related Tweet Classifiers," in *Proceedings of EMNLP*, 2024. [Online]. Available: https://aclanthology.org/2024.emnlp-main.34

[19] Z. S. Liu, R. Courant, and V. Kalogeiton, "FunnyNet-W: Multimodal Learning of Funny Moments in Videos in the Wild," *International Journal of Computer Vision*, vol. 132, no. 8, pp. 2885–2906, 2024. [Online]. Available: https://doi.org/10.1007/s11263-024-02000-2

[20] K. Ryokai, E. López, N. Howell, J. Gillick, and D. Bamman, "Capturing, representing, and interacting with laughter," 04 2018, pp. 1–12.

[21] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in *CVPR*, 2017.

[22] N. Takahashi and Y. Mitsufuji, "Multi-Scale multi-band densenets for audio source separation," in *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics*, vol. 2017-Octob, 2017, pp. 21–25.

[23] D. Niizumi, D. Takeuchi, Y. Ohishi, N. Harada, and K. Kashino, "BYOL for Audio: Exploring Pre-Trained General-Purpose Audio Representations," *IEEE/ACM Transactions on Audio Speech and Language Processing*, vol. 31, pp. 137–151, 2023.

[24] N. Calbucura, J. Guillen, and V. Barriere, "A Simple Method to Enhance Pre-trained Language Models with Speech Tokens for Classification," 4 2026. [Online]. Available: http://arxiv.org/abs/2512.07571

[25] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio set: An ontology and human-labeled dataset for audio events," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 776–780.

[26] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, "Fsd50k: An open dataset of human-labeled sound events," *IEEE/ACM Trans. Audio, Speech and Lang. Proc.*, vol. 30, p. 829–852, Dec. 2021. [Online]. Available: https://doi.org/10.1109/TASLP.2021.3133208

[27] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining*, 2008, pp. 413–422.

[28] V. Barriere, N. Gomez, L. Hemamou, S. Callejas, and B. Ravenet, "Standup4ai: A new multilingual dataset for humor detection in stand-up comedy videos," in *Findings of the Association for Computational Linguistics: EMNLP 2025*. Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 16 951–16 959.

[29] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio Set: An ontology and human-labeled dataset for audio events," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 776–780.

[30] A. Brown, V. Kalogeiton, and A. Zisserman, "Face , Body , Voice : Video Person-Clustering with Multiple Modalities," *ICCV Workshops*, pp. 3184–3194, 2021.

[31] A. Kuznetsova, "Multilingual Multimodal Detection of Humour in Stand-Up Comedy," Ph.D. dissertation, 2024. [Online]. Available: https://aclanthology.org/2024.lrec-main.1037/

[32] A. Clifton, S. Reddy, Y. Yu, A. Pappu, R. Rezapour, H. Bonab, M. Eskevich, G. J. Jones, J. Karlgren, B. Carterette, and R. Jones, "100,000 Podcasts: A Spoken English Document Corpus," *COLING 2020 - 28th International Conference on Computational Linguistics, Proceedings of the Conference*, pp. 5903–5917, 2020.

[33] Y. Gong, J. Yu, and J. Glass, "Vocalsound: a Dataset for Improving Human Vocal Sounds Recognition," *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*, vol. 2022-May, pp. 151–155, 2022.

[34] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2012. [Online]. Available: http://dl.acm.org/citation.cfm?id=2078195%5Cnhttp://arxiv.org/abs/1201.0490

[35] H. H. Wu, P. Seetharaman, K. Kumar, and J. P. Bello, "Wav2Clip: Learning Robust Audio Representations From Clip," *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*, vol. 2022-May, pp. 4563–4567, 2022.